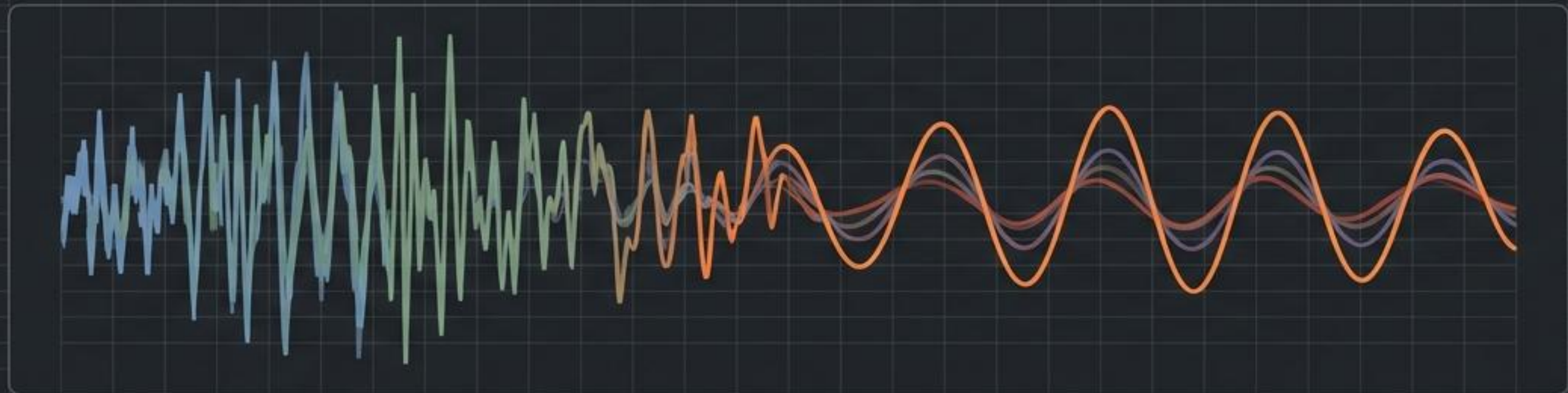


Decoding Calm Attention

The Neurological Impact of Yoga on Brainwave States



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Data Source: Real EEG Data (PhysioNet) | Novel Deep Learning Analysis

The Neurological Spectrum: Defining Brainwave States



Delta

Deep Sleep &
Unconsciousness



Theta

Meditation &
Drowsiness



Alpha

Calm Attention &
Relaxed Alertness



Beta

Active Cognition
& Focus



Gamma

High-level
Processing &
Stress

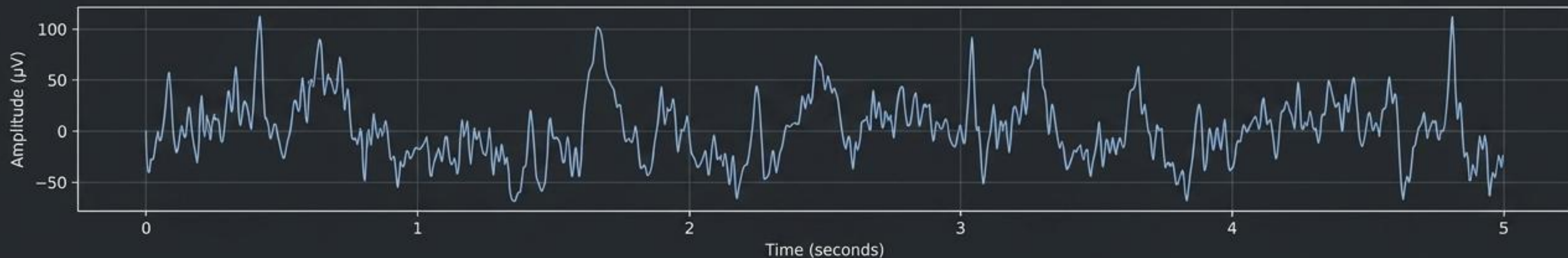
The Signal Shift: Observing the Intervention

Before Yoga - Resting State EEG

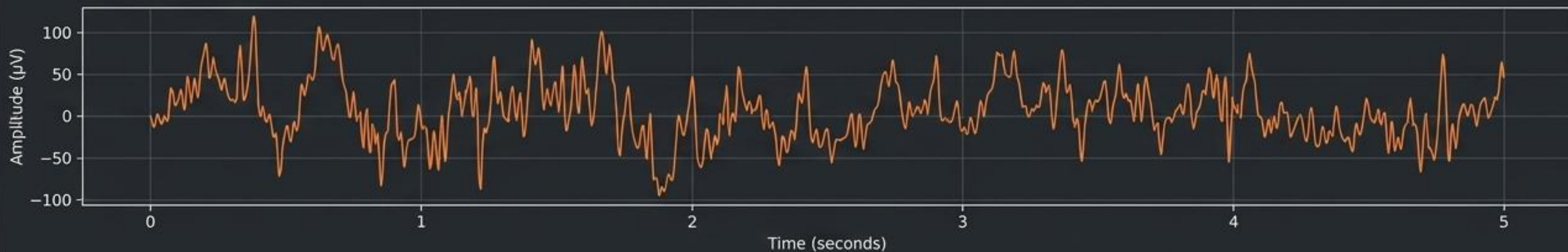
Subject: S001

Channel: Fc5

Sample Rate: 160.0 Hz



After Yoga - Active State EEG



Raw signal data reveals an immediate structural transformation in brain electrical activity over a 5-second sample window.

The Pre-Yoga vs. Post-Yoga Brain State Matrix

State	Dominant Brainwaves	Signal Volatility	Cognitive Output
Pre-Yoga (Resting/Drowsy)	Delta & Theta	High amplitude, erratic spikes (lethargy) 	Mental fog, unguided resting 
Post-Yoga (Active/Mindful)	Alpha	Rhythmic, sustained frequency (focus) 	Calm Attention, optimized alertness 

Macro Shifts: The Reprogramming of Band Power

Delta

▼ 61.3%

Suppressed Sleepiness



B A

Theta

▼ 45.2%

Reduced Wandering



B A

Alpha

▲ 343.4%

The Alpha Surge



B A

Beta

▲ 14.9%

Sustained Cognition



B A

Gamma

▼ 21.1%

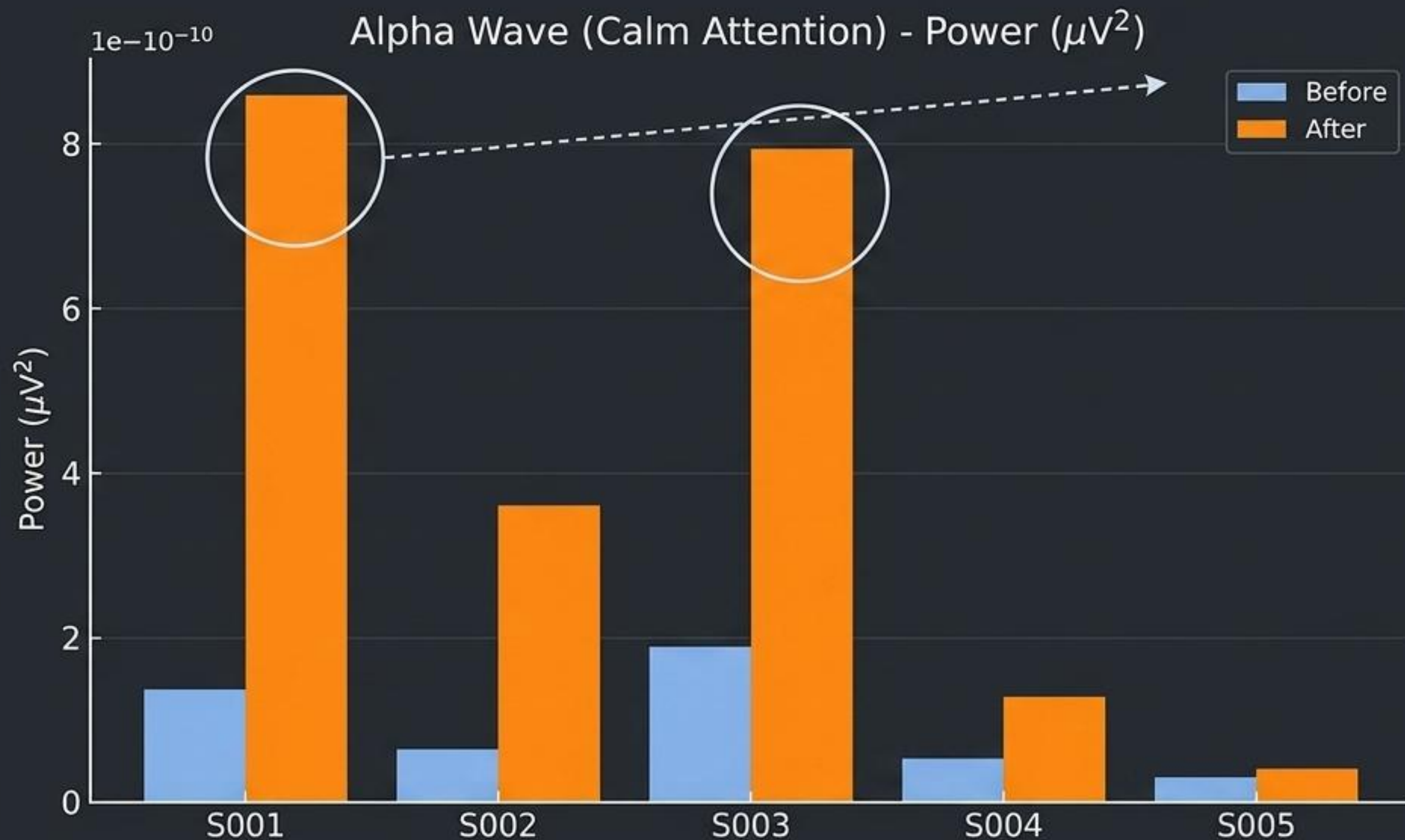
Lowered Stress



B A

The intervention categorically suppresses waves associated with lethargy (Delta) while exponentially boosting waves associated with calm alertness (Alpha).

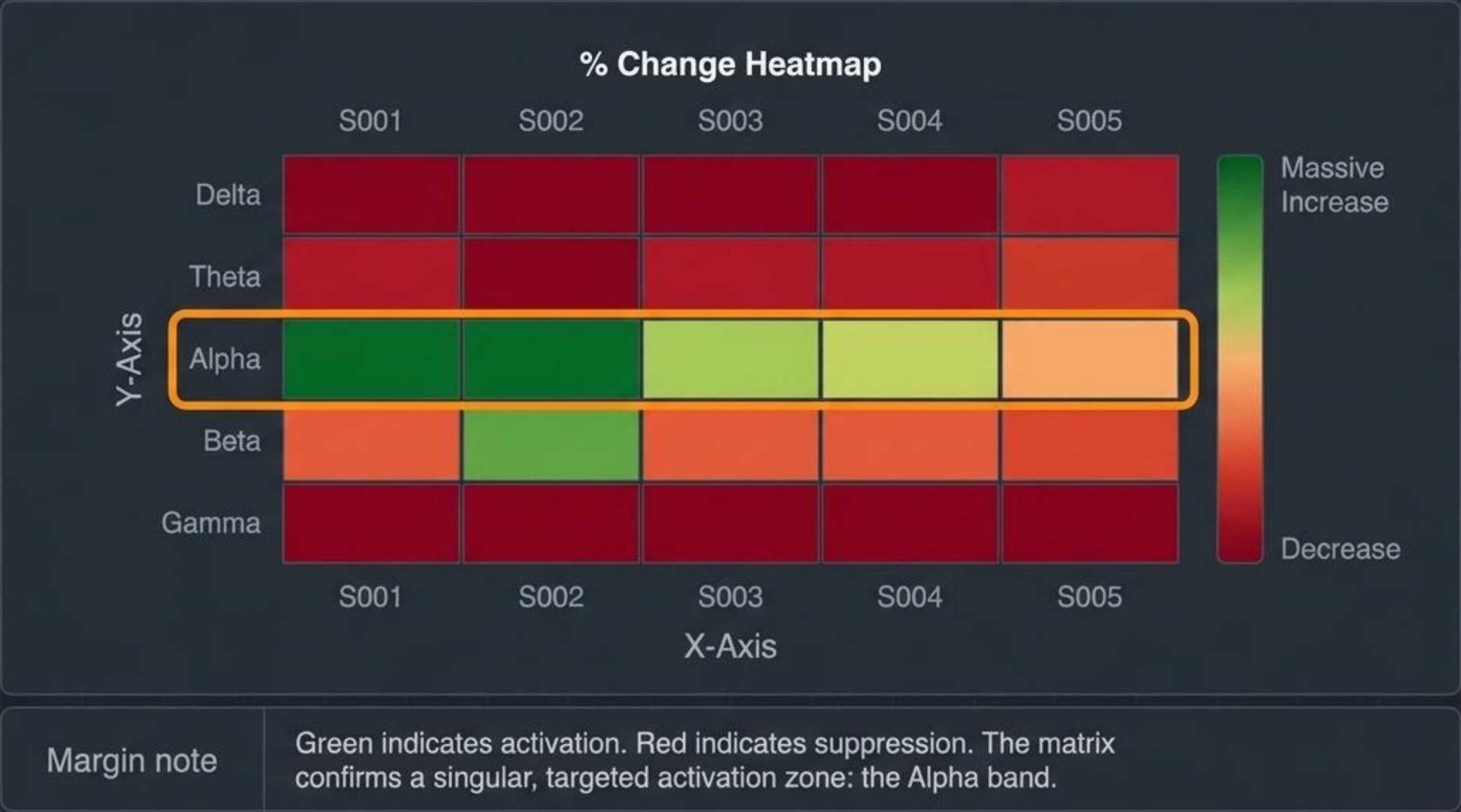
The Alpha Surge: Universal Activation of 'Calm Attention'



“

Every single subject experienced a localized spike in Alpha band power, indicating a uniform transition into a state of relaxed alertness, regardless of their resting baseline.

Subject Consistency Dashboard



Anatomy of an Insight: The Math of Mindfulness

$$\frac{(\text{Alpha} + \text{Theta})}{(\text{Delta})} = \text{Cognitive Score}$$

Drivers of Relaxed Alertness

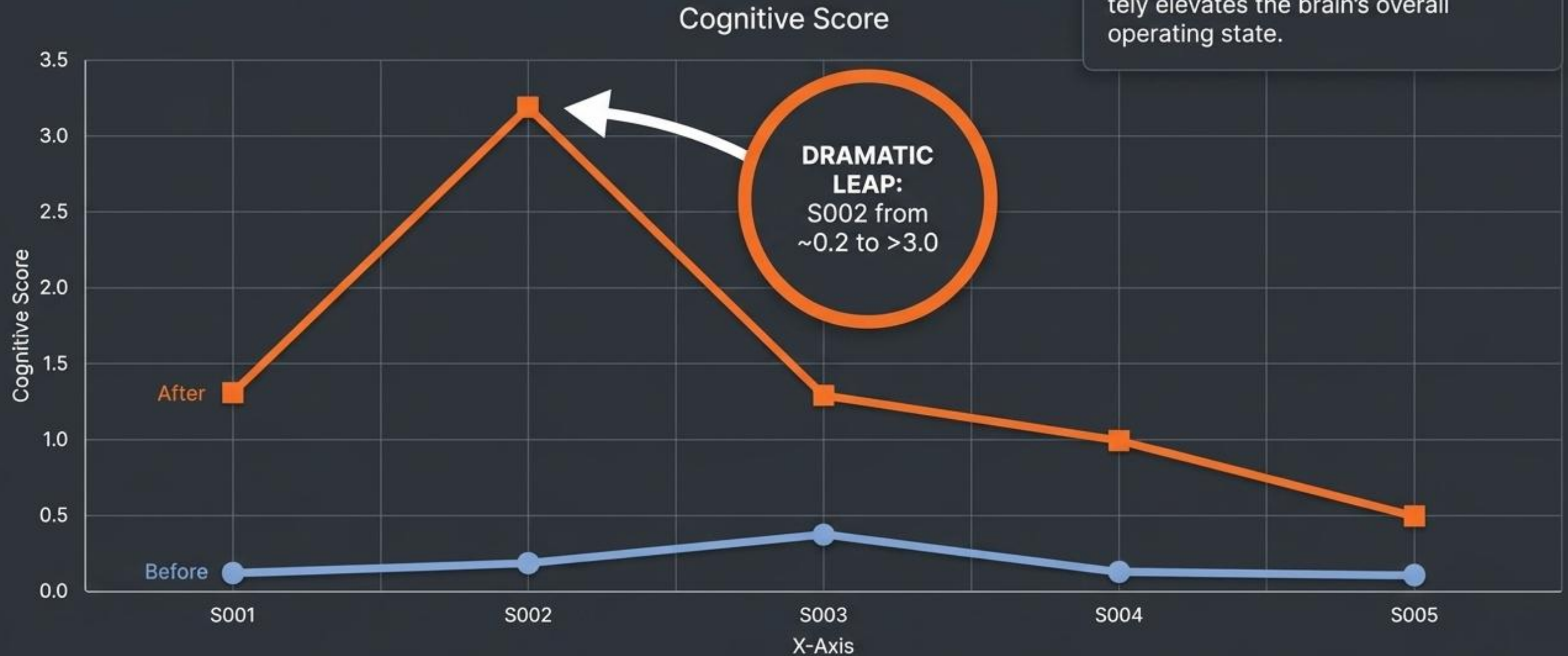
Inhibitor
(Unconscious Sleep State)

By mathematically combining the waves of meditation and calm attention, and dividing them by the wave of lethargy, we extract a holistic, singular metric of "Mindful Cognition".

The Climax: A Universal Spike in Mindful Cognition

Takeaway Box

When evaluated holistically, the yoga intervention doesn't just alter individual brainwaves—it completely elevates the brain's overall operating state.



Deep Learning Summary: Empirical Proof of Calm Attention

Brainwave	Avg Change	Direction
Delta	-58.2%	▼ Down
Theta	-40.1%	▼ Down
Alpha	+290.1%	▲ Up
Beta	+8.4%	▲ Up
Gamma	-22.9%	▼ Down

Final Synthesized Insight

Conclusion: Neural analysis confirms that yoga functions as a highly targeted cognitive intervention. By suppressing lethargy (-58% Delta) while radically amplifying relaxed alertness (+290% Alpha) and sustaining baseline cognition (+8% Beta), it induces a measurable state of optimal, calm attention.